

# **Pennsylvania Gaming Control Board**

## **Administrative Clarification Regarding Maximum Odds Regulation**

- The odds of any winning combination may not exceed 50 million to 1.
- The odds of achieving the highest advertised award that is based solely upon chance shall occur at least once in 50 million games.
- This does not apply to multiple awards won together in the same game where the aggregate prize is not advertised. However, it does consider the highest single multiplier (or advertised cumulative multiplier), as well as win-up-to statements.
- This odds rule shall not apply to games which make it possible for a player to win the highest advertised award multiple times through a bonus/feature.
- This rule shall apply to all wager categories that can win the highest advertised award.
- If the highest advertised award can occur within a bonus/feature, the odds calculation shall include the odds of obtaining the bonus/feature including the odds to achieve the award.
- The odds rule does not apply to purely informational cumulative win limit language, provided it is not prominently displayed. Usually, this information is contained in an “About” section at the end of the help screens. Such cumulative win limits must still not decrease the theoretical payback percentage by more than 0.001%.

### **Incrementing Multipliers (With or Without Limit)**

#### **Previously Methodology**

All increments of the multiplier were considered, regarding limitless incrementing multipliers non-compliant.

#### **Updated Methodology**

If a game has an incrementing multiplier, only the first increment of the multiplier is considered when determining the award values to be tracked, unless a higher increment is explicitly advertised.

### **Linked Odds**

The Board will continue to use the LINKED ODDS method instead of JURISDICTIONAL ODDS method. GLI will now adopt this method as well — this may allow some previous failures to become compliant. It will be up to the supplier

to decide if they want to resubmit any games that were previously determined to be non-compliant.

LINKED ODDS look at the total number hits for a winning combination across a set of plays, while JURISDICTIONAL ODDS instead look at the total number of plays with one or more hits for that particular winning combination. For example, if 100 million simulations were run, and 1 game had 2 hits of a particular winning combination while the other 99,999,999 games all had 0 hits of that particular winning combination, LINKED ODDS would be  $100,000,000/2=50,000,000$  (passing) but JURISDICTIONAL ODDS would be  $100,000,000/1=100,000,000$  (failing).

## **Multiple Multipliers**

### Previous Methodology

Wild Multipliers: Wild multipliers should be considered. If more than one wild cause an increase in multiplier, that increased multiplier should also be considered. For example, if a 5-reel game has wilds on reels 2-4 only, and each wild multiplies the win by 2x, we would consider 1x, 2x, 4x, and 8x multipliers as those are all possible multipliers. The odds are determined by combining the odds of the winning combination and the multiplier(s).

### Updated Methodology

Wild Multipliers / Multiple Multipliers: Wild multipliers should be considered. If it is possible to obtain more than one multiplier in a wager, all advertised multipliers should also be considered. If obtaining multiple multipliers results in an increase in multiplier, the increased multiplier value should be tracked.

- In cases where all possible multipliers are advertised to the player, then awards with all possible multipliers, including 1x, should be tracked.
- In cases where only some multipliers are advertised, we would consider at least one of the multipliers. For example, if a 5-reel game has wilds on reels 2-4 only, and each wild multiplies the win by 2x, we would track pays with a 1x multiplier (no multiplier), and then at least one increase due to the wild multiplier (getting at least one 2x multiplier). This applies regardless of whether the multiplier is multiplicative or additive.
- If a maximum multiplier is advertised, but not all possible multiplier combinations are advertised, we would consider all multipliers that are advertised. For example, if a 5-reel game has wilds on reels 2-4 only, and each wild multiplies the win by 2x, 3x, 4x, or 5x, and a maximum multiplier of 125x is advertised, we would track pays with a 1x multiplier (no multiplier), and then at least one increase due to the wild multiplier (getting at least one 2x multiplier, at least one 3x multiplier, at least one 4x multiplier, at least one 5x multiplier), as well as getting the maximum advertised multiplier of 125x. This is the same regardless of whether the multiplier is multiplicative or additive.

## **Win Up To Odds**

Any values that are advertised as “win up to” or similar need to meet the 1 in 50 million odds regulation.

## **Ranges of Values**

When a range of values is advertised, only the end points of the ranges need to be considered for odds evaluation. Any applicable multipliers should also be considered as outlined in the multiplier guidelines above.

## **Additional Advertised Award Clarifications**

When considering advertised awards for odds regulations, advertisements that occur after the wager should also be considered. The advertised awards are not limited to only what is in the help screens, and include splash screens, game banners, or other transitional media.

For example, if the help screens for a bonus feature advertise a range of values, but during the bonus all possible values are shown to the player, then all the awards advertised during the bonus should also be considered for odds tracking.

### **Task 1.**

A slot game uses a cluster payout system with a bonus feature where wild symbols lock onto the board, move across it, and increase their multiplier by +1 per step until they exit the grid. The game advertises a highest possible cluster win of 20×, formed by a combination of wilds ending at 5×, 5×, and 10× multipliers.

The task is to determine what must be tracked in a simulation or analytical model to ensure compliance with Pennsylvania Gaming Control Board maximum odds regulations, specifically that the highest advertised award occurs at least once in 50 million games.

This requires modelling and tracking the full probability chain, including: feature trigger probability, wild traversal and final multiplier outcomes, cluster formation probability, and the combined event leading to the advertised 20× payout under linked odds methodology.

### **Solution 1.**

Let  $W$  be the set of all cluster-win combinations (all cluster sizes  $\times$  all payout symbols). For every  $\omega \in W$ , track occurrences under the multipliers:

$$M = \{1 \times, 2 \times, 5 \times, 10 \times, 20 \times\}$$

where:

- $1 \times$  = default multiplier,

- $2 \times$  = first interment (per PGCB incrementing multiplier rule),
- $5 \times$  and  $10 \times$  = explicitly advertised multipliers,
- $20 \times$  = maximum advertised multiplier.

For each pair  $(\omega, m)$ , count occurrences using **Linked Odds**:

$$\text{Odds}(\omega, m) = \frac{N}{H(\omega, m)}$$

where:

- $N$  = total simulated games,
- $H(\omega, m)$  = total hits of combination  $\omega$  with multiplier  $m$ .

Compliance requires:

$$\text{Odds}(\omega, m) \leq 50,000,000$$

Or equivalently

$$P(\omega, m) \geq \frac{1}{50,000,000}$$

For every tracked advertised event.

## Direction of Regulatory Evolution in Modern Slot Probability Systems

Recent regulatory updates, particularly the adoption of **linked odds methodology**, the restriction to **first-increment multiplier tracking**, and the formalization of how **advertised awards are defined**, indicate not a relaxation of standards, but a **redefinition of what “probability of an advertised outcome” means in complex modern slot systems**.

Mathematically, the direction of evolution can be summarized as:

- From **state-explosion counting** (jurisdictional odds over all game outcomes)
- Toward **event-aggregation models** (linked odds over reward occurrences)

Formally, regulators are shifting evaluation from:

$$\mathbb{P}(\text{win event per game state})$$

to:

$$\mathbb{P}(\text{advertised reward occurrences across simulation space})$$

This does not lower the requirement:

$$\mathbb{P}(\text{max advertised award}) \geq \frac{1}{50,000,000}$$

but instead changes **how**  $\mathbb{P}$  is measured in systems with:

- Incrementing multipliers.
- Events formed from a combination of many more granular events.

## What this implies about the “direction of the wind”?

### 1. Not loosening, but abstraction

Regulations are not becoming easier; they are becoming **less sensitive to internal path complexity** and more focused on **observable payout events**.

### 2. Design freedom is increasing, verification burden is shifting

Developers gain flexibility in mechanics (wild movement, multipliers, chained bonuses), but must ensure that:

$\sum H(\text{advertised events})$  is still statistically bounded

### 3. Compliance is moving from combinatorics to simulation

Certification increasingly depends on:

- Monte Carlo linked-event counting
- empirical frequency of advertised outcomes
- correct inclusion of all advertised multiplier states

The “wind” is not toward looser regulation, but toward a **standardized interpretation of what counts as a measurable gambling event in highly compound systems**.

In essence:

The regulator is no longer trying to count every possible path in the forest — only how often the advertised tree is actually reached.